

Mattia Domizio

Computer Engineer | Software Engineer | Data Engineer

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🇪🇺 EU Citizenship (Italian)

SUMMARY

Computer Engineer, MSc in Computer Engineering (**Software track**) from Politecnico di Torino (**110/110 cum laude**), with hands-on experience across **full-stack web development**, **distributed systems**, **data engineering** and **low-level system and GPU programming**. The Software track shaped a strong **DevOps/DevSecOps**, **Cloud Native** and **Kubernetes** background, delivering across both **Waterfall** and **Agile/Scrum** methodologies with close attention to **semantic versioning**, **well-defined APIs** and **SLOs/SLAs**. Completed an **external thesis and internship at Target Reply** building **BI-as-code** tooling on top of a metadata governance platform. Looking for a full-time role as **Software Engineer**, with a solid foundation in **Data Engineering** and **AI Engineering**, where I can ship reliable systems and keep learning.

EDUCATION

MSc in Computer Engineering (LM-32): Software Track <i>Politecnico di Torino</i> 110/110 cum laude (graduated July 24, 2026)	2024 – 07/2026
BSc in Computer Engineering (L-08) <i>Politecnico di Torino</i> 107/110	2021 – 2024
Scientific High School Diploma <i>Liceo Scientifico Cattaneo, Turin</i> 100/100	2021
Cambridge B2 First Certificate <i>English language certification</i>	2020

WORK EXPERIENCE

Data Engineer Intern: External Thesis Project <i>Target Reply S.r.l.</i> • Took part in the development of a metadata governance platform as part of the master's thesis project, building BI-as-code tooling on top of DataHub metadata. • Designed and implemented an MCP server generating versionable Power BI semantic models/reports (PBIP) and NEO, a cross-platform BI frontend orchestrator (see Projects).	02/2026 – 07/2026
University Tutor <i>Politecnico di Torino</i> • Supported undergraduate students with lab exercises and code reviews on the OOP course (Java).	02/2025 – 07/2025
Private Tutor <i>Academic support and homework assistance</i>	2020 – 2024

SKILLS

Languages

Python, Rust, C, C++, Java, Kotlin, TypeScript, JavaScript, Bash, Powershell, GDScript, CUDA, SQL, HTML5, CSS3, Assembly Script (ARM, x86-64, RISC-V, 32), Scala, LaTeX

Frontend & UI

React, React Router, Redux Toolkit, Vite, Next.js, Bootstrap, TailwindCSS, Qt

Backend & APIs

Node.js, Express, Flask, Django, OpenAPI / Swagger, REST, JWT, Postman, .NET, Spring, Spring Boot, Model Context Protocol (MCP), GraphQL, gRPC, HTTP/2, WebSockets, SSE, OAuth2 / OIDC (Keycloak), MQTT, Kafka, PayPal API

Data & Databases

PostgreSQL, MySQL, SQLite, CloudNativePG, MongoDB, Redis, Apache Hadoop, Apache Spark, DataHub, Debezium (CDC), Power BI, Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, Databricks, Snowflake, dbt, Elasticsearch, Qlik, MicroStrategy

Cloud & DevOps

Docker, Kubernetes, Argo CD (GitOps), Strimzi, Git, GitHub, GitLab, GitHub Actions, AWS, Azure, CMake, Helm, Gradle, NPM, SonarQube

Game Dev & Design

Godot Engine, Figma, GIMP, NVIDIA tooling

OS & Tooling

Linux, Windows, Arch, Nix, WireGuard

ADDITIONAL

Languages

- Italian (Native)
- English (**B2 First Cambridge**)

Certifications & Events

- **SAS Hackathon Boot Camp**, Milan, 2026 (SAS Innovate on Tour): completed hands-on data & AI challenges on **SAS Viya** in the SAS Cloud (Azure), spanning the full data and AI life cycle.

PROJECTS

PBIP MCP Server: Power BI BI-as-code Generator

2026

Model Context Protocol (MCP), Python, Power BI (PBIP), JSON Schema, DataHub, LLM Agents

- Built an **MCP server** that generates versionable Power BI semantic models and reports in **PBIP** format, implementing the **BI-as-code** paradigm.
- Supports three ingestion paths: a **JSON IR** produced by external frontend tooling, an existing PBIP project (parsed, modified and validated against official **JSON Schemas** with custom consistency checks), or metadata ingested into **DataHub**.
- Implemented an **agentic** component that inspects business KPIs and resolves them against DataHub's metadata context to automatically populate the semantic model.
- Developed at **Target Reply** as part of the master's thesis project.

NEO: BI Frontend Orchestrator

2026

Model Context Protocol (MCP), Python, Power BI, MicroStrategy, Qlik, BI-as-code

- Designed an orchestrator that converts BI/data frontends into a **BI-as-code** paradigm, making them versionable and automatable.
- Implemented automatic **cross-platform migrations** between BI technologies (currently **MicroStrategy**, **Qlik** and **Power BI**), with an architecture built for continuous extension to new platforms.
- Built custom **MCP tools** for day-to-day governance and operation of the supported frontend platforms (creating/editing dashboards, auditing and enforcing standards).
- Developed at **Target Reply** as part of the master's thesis project.

EuroTransit: Cloud-Native Train Ticket Marketplace

2026

Kubernetes, Helm, Docker, Argo CD (GitOps), Kotlin, Spring Boot, Kafka, Strimzi, Debezium, CloudNativePG, React, Vite, TypeScript

- Team capstone project (4 members) for **Cloud Programming and Operations** at Politecnico di Torino: a Kubernetes-native train-ticket marketplace (browsing, booking, seat reservation, PayPal payments) proving operational behaviour under partial failure.

- Implemented the **money path** as a fully asynchronous, orchestrated **Kafka** saga (Kotlin/Spring Boot) with a **transactional outbox** drained by **Debezium** CDC, atomic never-oversell seat reservation and idempotent, exactly-once payment authorization/capture.
- Structured as a two-repository **GitOps** model (application + configuration) continuously reconciled by **Argo CD** on **CloudNativePG**-backed, Helm-templated services with progressive delivery (canary, blue/green).
- Built the **React / Vite / TypeScript** storefront with Keycloak OAuth2 PKCE auth and live saga tracking over SSE.

Godot Shader Pipeline Warm-Up

2025 – 2026

Godot Engine, GDScript, GDShader, Graphics Pipeline, Benchmarking

- Designed and implemented a **deterministic shader pipeline warm-up workflow** in Godot to reduce first-use stutter at runtime.
- Ran benchmarks comparing cold vs. warmed-up runs and documented results in a dedicated **report.pdf**.
- **GitHub**: [link](#)

Participium: Civic Reporting Platform

2026

Full-stack Web (TypeScript, Python, React, JavaScript, HTML, CSS), Software Engineering (Swagger / OpenAPI, Docker, Bash, Redis), Agile (YouTrack)

- Civic reporting platform for the city of Turin: citizens submit reports on urban issues and municipal officers triage and manage them.
- Exam project for **Software Engineering II** at Politecnico di Torino.
- **GitHub**: [link](#)

GeoControl

2025

REST APIs, TypeScript, React, JavaScript, HTML, CSS, Docker, Swagger / OpenAPI

- Monitoring system for heterogeneous sensors (position, temperature, pressure, humidity, gas concentration and other geo-environmental variables).
- Designed the REST API surface and the underlying data model; full documentation shipped under the `/doc` folder.
- **GitHub**: [link](#)

Ruggine: Rust Chat App

2025

Rust, Tokio, WebSockets, React, JavaScript, HTML, CSS

- Built a **secure messaging system** in Rust with private and group chats.
- Exam project for **System and Device Programming** (*Programmazione di Sistema* course) at Politecnico di Torino.
- **GitHub**: [link](#)

Stuff Happens

2025

React, Node.js, Express, SQLite, JWT, JavaScript, HTML, CSS

- Web-based card game where players guess the position of a new “bad luck” card relative to three known cards, scoring on a custom *bad luck index*.
- Features **user authentication**, game history tracking and a travel-mishap themed deck (“Unlucky Holidays”).
- **GitHub**: [link](#)

Film Manager Service

2024 – 2025

REST APIs, Swagger / OpenAPI, JSON Schema, JavaScript, Postman

- REST service to manage a film catalogue, with an **OpenAPI/Swagger** specification, **JSON Schemas** for payload validation and a Postman collection covering the main flows.
- Exam Call 1 project (**Distributed Systems**).
- **GitHub**: [link](#)

PAC-MAN on LPC1768

2023 – 2024

ARM Assembly, C, Embedded C

- Ported PAC-MAN to the **LPC1768** board using ARM **assembly** and C, targeting the on-board peripherals.
- Exam project for **Computer Architectures** at Politecnico di Torino.
- **GitHub**: [link](#)

OTHER EXPERIENCES

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- **Weekend Fruit & Vegetable Clerk** - Gabrielis S.r.l. (11/2023 – 02/2024)
 - **Election Poll Worker** - Comune di Torino (09/2022)
 - **Assistant Coach** - A.S.D. Pozzomaina (10/2017 – 06/2019)